AI101 MINI-PROJECT | TEAM [no.]

Basketball Match Winner Predictor

Predicting which team wins a match with Logistic Regression

Repo: github.com/Mayur-Ukey-358/AI101-Project

TEAM AND ROLES

Mayur Ukey

Documentation lead

[Name]

Execution lead

[Name]

PPT lead

[Name]

Video lead

Problem and data

WHAT AND WHY

Predict which of two teams wins a match from pre-match team statistics: ratings, recent form, rest and home advantage. It shows which factors drive wins and how far a simple model can go.

10,000	22	51% / 49%
matches (rows)	columns	class 0 / class 1

DATASET SOURCE

Kaggle: Basketball Tournament Dataset (sharmajicoder). It is a simulated dataset, so the results show that the method works, not how real games would turn out.

COLUMNS

- Per team (prefix A_ and B_): Off, Def and Net rating, Star Player, Chemistry, Injury, Last5 WinPct, Momentum, Rest
- Match: Home_Team, Tournament_Stage
- Target: Winner
- Dropped: Match_ID (identifier only), leaving 20 input features

TARGET

Winner: 1 = Team A wins, 0 = Team B wins. A yes/no outcome, so we use logistic regression.

Approach: data → model → output

1 Data

synthetic_basketball_dataset.csv, 10,000 × 22

data/

2 Prepare

Drop Match_ID, encode Home_Team, fill gaps

data_prep.py

3 Split

80% training, 20% testing (stratified)

data_prep.py

4 Model

Pipeline: scaling + Logistic Regression

train_model.py

5 Output

Metrics, graphs, match predictions

evaluate.py, predict.py

LIBRARIES USED

pandas **numpy** **scikit-learn** **matplotlib** **seaborn**

Run everything with: python src/main.py

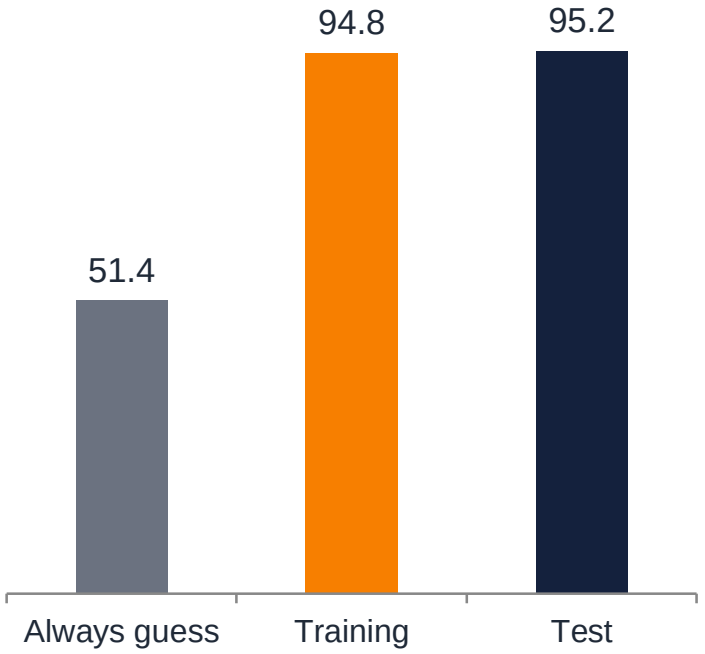
Results

95.2%	0.946	0.955	0.95
Accuracy	Precision	Recall	F1-score

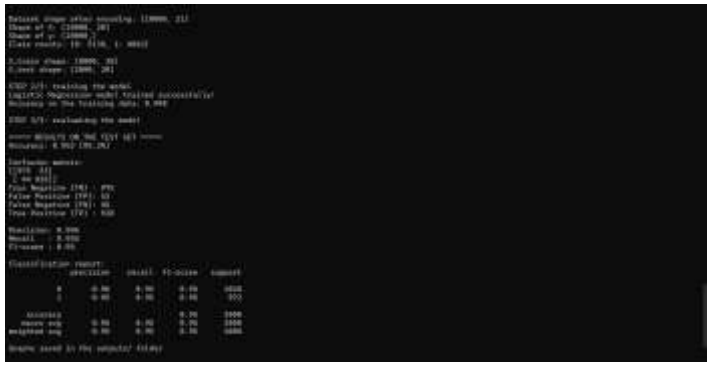
CONFUSION MATRIX (2,000 TEST MATCHES)

	Predicted 0	Predicted 1
Actual 0	975 TN	53 FP
Actual 1	44 FN	928 TP

ACCURACY VS GUESSING (%)



DEMO RUN: python src/main.py



Test accuracy 95.2% against 51.4% for always guessing the more common outcome. Training and test accuracy match, so no overfitting.

Learnings

WHAT WORKED

- Scaling plus logistic regression reaches 95.2%, far above 51.4% for guessing
- Training 94.8% and test 95.2%: no overfitting
- Errors are balanced: 53 false positives, 44 false negatives

WHAT DIDN'T / LIMITS

- Data is simulated, not real games
- Net Rating = Off − Def, so those features overlap
- Only a linear model, no tuning or other models tried

WHO DID WHAT

- Documentation: data_prep.py, README
- Execution: train_model.py, main.py, requirements.txt
- PPT: evaluate.py, slides
- Video: predict.py, video

AI-USE DISCLOSURE Claude (Anthropic) helped write and debug the code and draft the README and slides. We ran and tested every file ourselves, and each member can explain their part. All results come from running our code on the Kaggle dataset.